

CONFIDENTIAL



UNIVERSITI TEKNOLOGI MALAYSIA FACULTY OF COMPUTING

TEST 2 (PART 1)

SEMESTER I 2021/2022

SUBJECT CODE : SECJ/SCSJ2013
SUBJECT NAME : DATA STRUCTURES AND ALGORITHMS
YEAR/COURSE : 2 (SCSJ/ SCSV/ SCSB/ SCSR/ SCSP)
TIME : 20.10 – 21.20 MYT (1 hour 10 minutes)
DATE : 12th JANUARY 2022 (WEDNESDAY)

INSTRUCTIONS TO THE STUDENTS:

- Read the problem and instructions carefully.
- You are given **ONE HOUR TEN MINUTES** to complete the test inclusive of the submission of your program (**1 hour to answer** the question, **10 minutes to submit** the answer).
- Your program must follow the input and output as required in the text and shown in the examples. You must test the programs with (but not limited to) all the input given in the examples.

IMPORTANT NOTES:

- All the **COMMENT STATEMENTS** in the submitted program **WILL NOT BE EVALUATED**.

SUBMISSION PROCEDURE:

- Only the source code is required for the submission and the source code's file shall be named as follows: ***Name_matricesNo_section.cpp*** (i.e. *AinaAli_A20EC018_01.cpp*).
- You do not need to compress the file.
- Submit the source code file via the **UTM's e-learning system**.

Question**[50 Marks]**

Complete the linked list program given in **Program 1** by writing the implementation for all of the **List** class's member functions and completing the **main** function to accomplish each of the following tasks:

```
1  //Program 1
2  class FruitNode {
3      private:
4          string name;
5          float price;
6
7      public:
8          FruitNode *next;
9          FruitNode(string n, float p) {
10              name = n;
11              price = p; }
12
13          void getFruitInfo() {
14              cout << name << " = RM" << price; }
15  };
16
17  class List {
18      private:
19          FruitNode *headNode;
20
21      public:
22          List() {
23              headNode = NULL; }
24
25          void task1(string, float);
26          void task2();
27          void task3(int, string, float);
28          void task4();
29  };
30
31  int main()
32  {
33      List fruitList;
34
35      //Incomplete main function definition
36
37      return 0;
38  }
```

Task 1: Define a function named **task1**. The function should be able to add a new fruit node to both an empty and non-empty list. The new node will be added to the front of the list if the list is not empty. **Figure 1** depicts the outcome of this task after several function calls were executed. **(10.5 marks)**

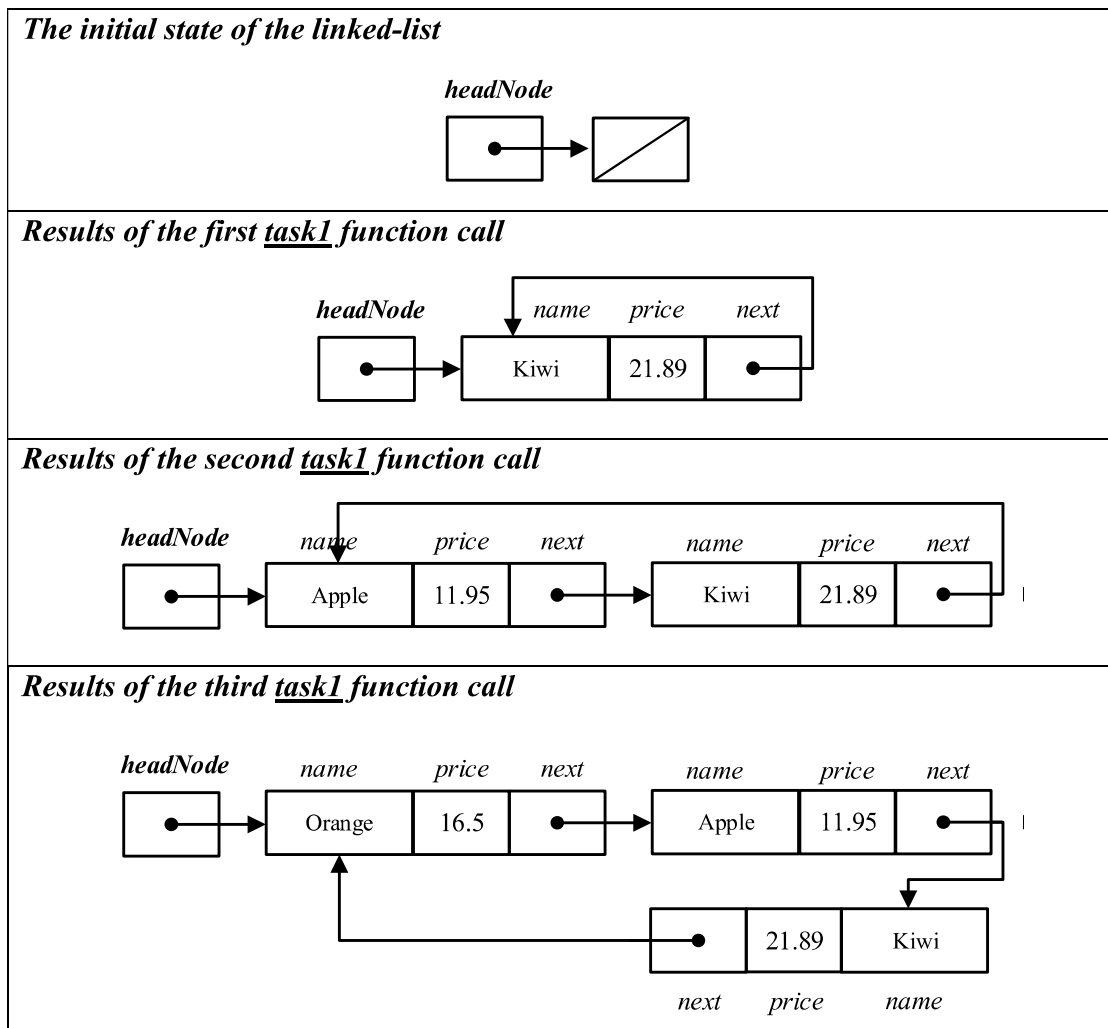


Figure 1: The latest nodes in the list after a few node insertions

Task 2: Define a function named **task2**. The function should be able to print all of the data in the list. **Figure 1** shows a sample output of this task after executing the **task2** function. In this task, you should use a loop and a selection. **(9.5 marks)**

```
Fruit List
Orange = RM16.50 >> Apple = RM11.95 >> Kiwi = RM21.89
```

Figure 2: Sample output after executing the **task2** function

Task 3: Define a function named **task3**. The function should be able to insert a new fruit node at a specific location. The value of the first argument passed to this function determines the position of the node that will be inserted. Assume the value of the first argument to this function is 2. This means that the node that will be inserted will become the second node in the list. **Figure 3** depicts the outcome of this task after executing the **task3** function with 2 as the first argument value. In this task, you should use a loop.

(9.5 marks)

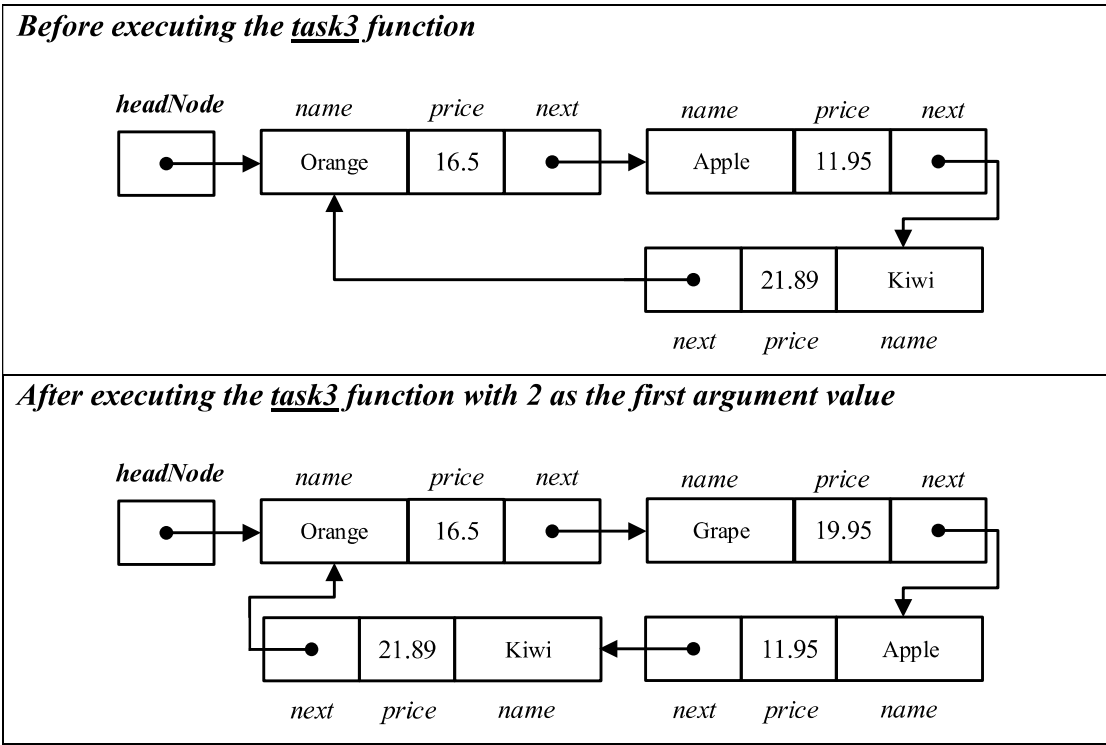


Figure 3: Current nodes in the list after inserting a new fruit node into the list

Task 4: Define a function named **task4**. The function should be able to remove the last fruit node from the list. **Figure 4** depicts the outcome of this task after a number of function calls were made.

(6.5 marks)

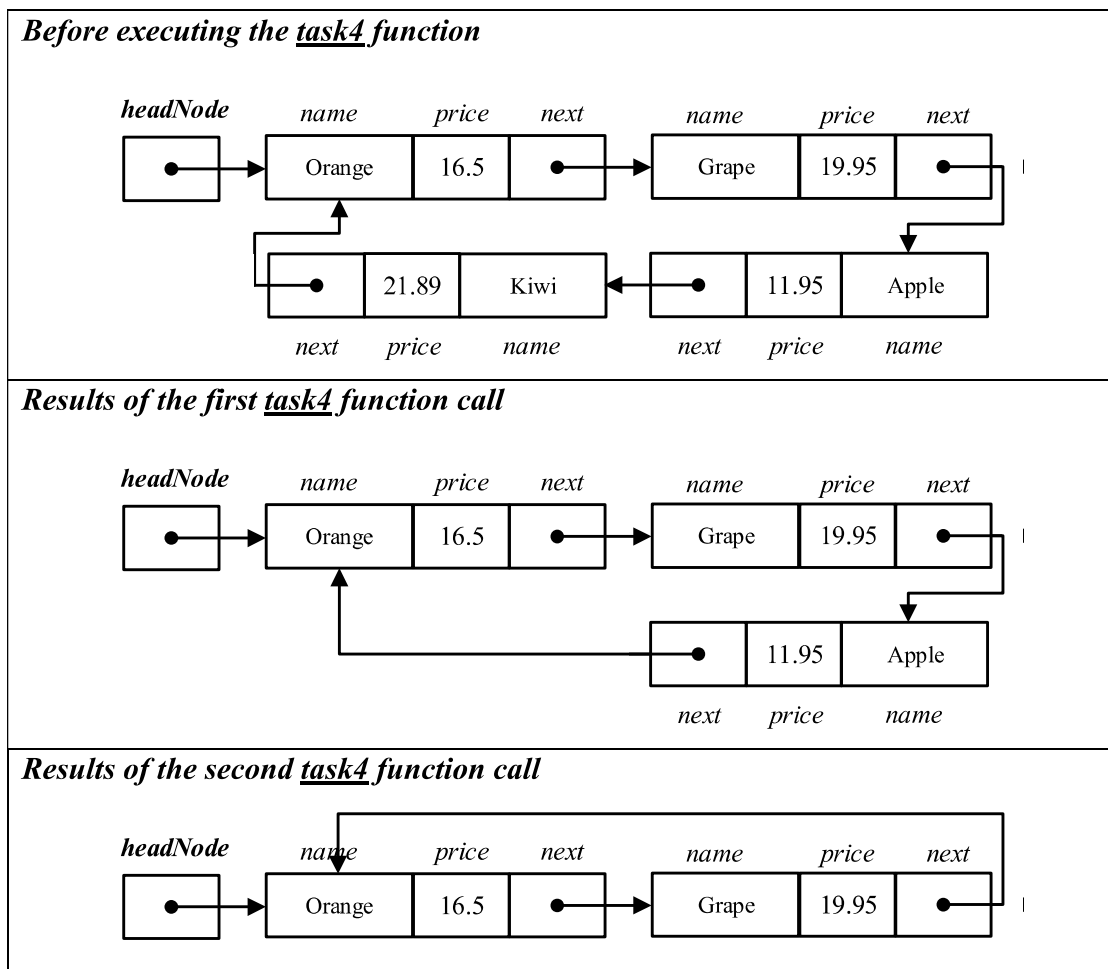


Figure 4: Current nodes in the list after the deletion of the last node in the list

Task 5: Complete the `main` function with a few function calls. Please use the appropriate function that has been defined and the appropriate argument to obtain the output shown in **Figure 5**. (14 marks)

```
Fruit List
Kiwi = RM21.89

Fruit List
Apple = RM11.95 >> Kiwi = RM21.89

Fruit List
Orange = RM16.50 >> Apple = RM11.95 >> Kiwi = RM21.89

Fruit List
Orange = RM16.50 >> Grape = RM19.95 >> Apple = RM11.95 >> Kiwi = RM21.89

Fruit List
Orange = RM16.50 >> Grape = RM19.95 >> Apple = RM11.95

Fruit List
Orange = RM16.50 >> Grape = RM19.95
```

Figure 5: The output of the completed Program 1